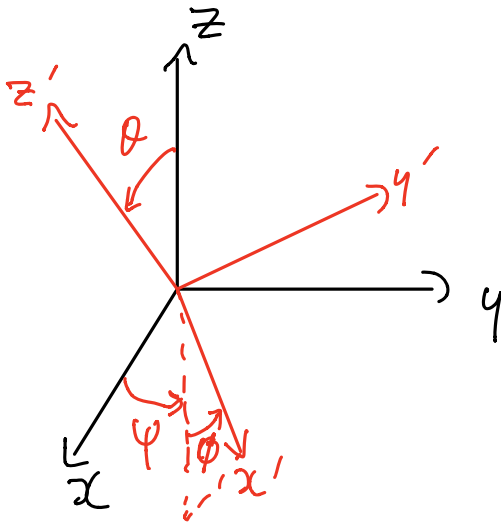


## I - Définitions

### 1) Angles d'Euler



$\theta$  : nutation

$\psi$  : précession

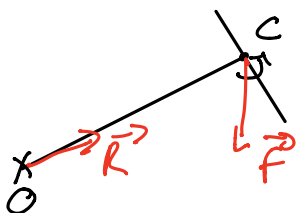
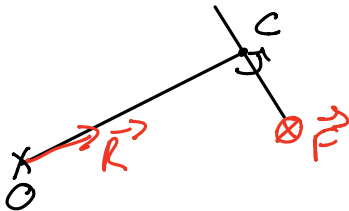
$\phi$  : rotation propre

### 2) Approximation gyroscopique

$$\vec{\omega} = \Omega \vec{e}_z$$

$$|\dot{\phi}| \gg \begin{cases} |\dot{\theta}| \\ |\dot{\psi}| \end{cases}$$

### 3) Mouvement du gyroscope



Thm moment cinétique :

$$\frac{d\vec{L}_O}{dt} = \vec{OC} \times \vec{F}$$

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$$\frac{d\vec{L}_O}{dt} = \frac{\vec{L}_O}{|\vec{L}_O|} \wedge \vec{\omega}_g$$